

One-dimensional Heat flow:-

1) Solve by the method of separation of variables, solve the Pdeqn

$$\frac{\partial u}{\partial t} = c^2 \cdot \frac{\partial^2 u}{\partial x^2}$$

Sol Given $\frac{\partial u}{\partial t} = c^2 \cdot \frac{\partial^2 u}{\partial x^2}$ — (1)

let $u = XT$ — (2)

diff (2) w.r.t t Partially

$$\frac{\partial u}{\partial t} = XT'$$

$$\frac{\partial^2 y}{\partial v^2} = T X''$$

Putting this value in ①

$$X T' = c^2 X'' T$$

$$\Rightarrow \frac{T'}{c^2 T} = \frac{X''}{X} = K \quad \text{--- (2)}$$

$$\Rightarrow \frac{X''}{X} = K \quad \text{and} \quad \frac{T'}{c^2 T} = K$$

$$\Rightarrow X'' - K X = 0$$

$$T' - c^2 T K = 0 \quad \text{--- (3)}$$

Case ① when $K = 0$

from ③

$$X'' = 0$$

$$\frac{d^2 X}{dv^2} = 0$$

$$m = 0, 0$$

$$X = C_1 v + C_2$$

$$T' = 0$$

$$\frac{dT}{dt} = 0$$

$$T = C_3$$

$$y(x, t) = X T = (C_1 v + C_2) \cdot C_3 \quad \text{--- (4)}$$

② when K is +ve, i.e., $K = p^2$
from ③ $T' - p^2 c^2 T = 0$

$$\Rightarrow X'' - b^2 X = 0$$

$$\Rightarrow (D^2 - b^2) X = 0$$

$$\Rightarrow m^2 - b^2 = 0$$

$$\Rightarrow (m+b)(m-b) = 0$$

$$\Rightarrow m = b, -b$$

$$X = c_1 e^{bx} + c_2 e^{-bx}$$

$$T' - b^2 c^2 T = 0$$

$$(D_1 - b^2 c^2) T = 0$$

$$m - b^2 c^2 = 0$$

$$m = \pm b^2 c^2$$

$$T = c_3 e^{c^2 b^2 t}$$

$$\text{Hence } u = XT$$

$$= (c_1 e^{bx} + c_2 e^{-bx}) \cdot c_3 e^{c^2 b^2 t} \quad \text{--- (5)}$$

Case (3) when K is -ve i.e. $K = -b^2$

$$X'' + b^2 X = 0$$

$$m^2 + b^2 = 0$$

$$m = \pm bi$$

$$X = c_1 \cos bx + c_2 \sin bx$$

$$T' + b^2 c^2 T = 0$$

$$m + b^2 c^2 = 0$$

$$m = -b^2 c^2$$

$$T = c_3 e^{-b^2 c^2 t}$$

$$\Rightarrow u(x, t) = XT$$

$$= (c_1 \cos bx + c_2 \sin bx) \cdot c_3 e^{-b^2 c^2 t}$$

$$\text{--- (6)}$$

Since u decreases as time t increases,
the only suitable soln is

$$(u(x, t) \rightarrow 0 \text{ when } t \rightarrow \infty)$$

$$u(x, t) = (c_1 \cos bx + c_2 \sin bx) e^{-b^2 c^2 t} \quad \text{--- (7)}$$

Now we apply boundary conditions.

$$u(0, t) = 0 \quad \& \quad u(l, t) = 0$$

$$\Rightarrow 0 = (c_1 + 0) e^{-b^2 c^2 t}$$

$$c_1 c_3 e^{-b^2 c^2 t} = 0.$$

$$\Rightarrow c_3 \neq 0 \quad c_1 = 0.$$

$$\text{from (7)} \quad u = c_2 \sin bx \cdot c_3 e^{-b^2 c^2 t} \quad \text{--- (8)}$$

$$\Rightarrow 0 = c_2 \sin bx \cdot c_3 e^{-b^2 c^2 t}$$

$$c_2 c_3 \neq 0.$$

$$\Rightarrow \sin bx = 0$$

$$\sin bx = \sin n\pi$$

$$bx = n\pi$$

$$b = \frac{n\pi}{l}$$

put this in (8)

$$u(x, t) = C_2 C_3 \sin \frac{n\pi x}{l} \cdot e^{-b^2 c^2 t}$$

$$\text{let } C_2 C_3 = b_n$$

$$\Rightarrow u(x, t) = b_n \sin \frac{n\pi x}{l} e^{-b^2 c^2 t}$$

hence most-general soln is

$$u(x, t) = \sum_{n=1}^{\infty} b_n \sin \frac{n\pi x}{l} \cdot e^{-b^2 c^2 t} \quad \text{--- (9)}$$

Now use initial conditions

$$u(x, 0) = f(x).$$

$$\Rightarrow \text{put } t = 0$$

$$f(x) = \sum_{n=1}^{\infty} b_n \sin \frac{n\pi x}{l}$$

$$b_n = \frac{2}{l} \int_0^l f(x) \sin \frac{n\pi x}{l} dx$$

Hence Eqn (9) is the required soln of heat-Eqn. where

$$b_n = \frac{2}{l} \int_0^l f(x) \cdot \sin \frac{n\pi x}{l} dx.$$

Problems based on Heat Equation

1) A rod of length l with insulated sides initially at a uniform temp 4_0 . its ends are suddenly cooled to 0°C and are kept at that temp. Find the temp $\theta(x, t)$.

Sol we have 1-D Heat eqn $\uparrow \theta(x, t)$

$$\frac{\partial \theta}{\partial t} = c^2 \frac{\partial^2 \theta}{\partial x^2} \quad \text{--- (1)} \quad \begin{array}{c} x=0 \longleftarrow l \longrightarrow x=l \\ \theta(x, 0) = 4_0 \end{array}$$

Boundary conditions: —

$$\theta(0, t) = 0, \quad \theta(l, t) = 0 \quad \forall t \quad \text{--- (2)}$$

$$\text{initial condition} \rightarrow \theta(x, 0) = 4_0 \quad \text{--- (3)}$$

Let $u = XT$ be the soln of eqn (1)

$$\Rightarrow XT' = c^2 X''$$

$$\Rightarrow \frac{X''}{X} = \frac{1}{c^2} \cdot \frac{T'}{T} = K \text{ (say)}$$

$$\Rightarrow X'' - KX = 0$$

$$T' - Kc^2 T = 0$$

$$\text{--- (4)}$$

$$\Rightarrow \text{--- (5)}$$

Case (1) when $K = 0$

$$X'' = 0$$

$$X = C_1 x + C_2$$

$$T' = 0$$

$$T = C_3$$

$$\therefore \theta(x, t) = XT = (C_1 x + C_2) C_3 \quad \text{--- (5)}$$

Case (2) when K is +ve i.e. $K = b^2$

$$\begin{array}{l|l}
 X'' - b^2 X = 0 & T'' - b^2 c^2 T = 0 \\
 \Rightarrow (D^2 - b^2) X = 0 & (D^2 - b^2 c^2) T = 0 \\
 \Rightarrow m^2 - b^2 = 0 & m - b^2 c^2 T = 0 \\
 \Rightarrow m = \pm b & m = b^2 c^2 T \\
 \Rightarrow X = C_1 e^{bn} + C_2 e^{-bn} & T = C_3 e^{b^2 c^2 T}
 \end{array}$$

$\therefore u(x, t) = X T$

$= (C_1 e^{bn} + C_2 e^{-bn}) (C_3 e^{b^2 c^2 T})$ (6)

Case (3) when K is -ve, i.e. $K = -b^2$

$$\begin{array}{l|l}
 X'' + b^2 X = 0 & T'' + b^2 c^2 T = 0 \\
 \Rightarrow (D^2 + b^2) X = 0 & (D^2 + b^2 c^2) T = 0 \\
 \Rightarrow m^2 + b^2 = 0 & m = \pm b^2 c^2 T \\
 m = \pm bi & T = C_3 e^{-b^2 c^2 T}
 \end{array}$$

$X = (C_1 \cos bn + C_2 \sin bn)$

$u = X T$

$= (C_1 \cos bn + C_2 \sin bn) \cdot C_3 e^{-b^2 c^2 T}$ (7)

(when $t \rightarrow \infty$, $u(x, t) = 0$ which satisfies this, that will be the most-general soln).

(5) is independent of t . (So not possible)
(6) as $t \rightarrow \infty$, $u \rightarrow \infty$ (" " " "
hence we have to take eqn (7)

$$\therefore u(x, t) = (C_1 \cos bx + C_2 \sin bx) \cdot C_3 e^{-b^2 c^2 t}$$

Now apply boundary conditions

$$u(0, t) = 0, \quad u(l, t) = 0 \quad \forall t$$

Put $x = 0$ in eqn (7)

$$0 = C_1 C_3 e^{-b^2 c^2 t}$$

if $C_3 = 0$ then eqn (7) is zero.
then $C_1 = 0$.

using in (7)

$$u = C_2 \sin bx \cdot C_3 e^{-b^2 c^2 t} \quad \text{--- (8)}$$

Put $x = l$ in eqn (8)

$$0 = C_2 C_3 \sin bl \cdot e^{-b^2 c^2 t} \quad (C_2 C_3 \neq 0)$$

$$\sin bl = 0$$

$$\sin bl = \sin n\pi$$

$$bl = n\pi$$

$$b = \frac{n\pi}{l}$$

$$\Rightarrow u(x, t) = C_2 C_3 \sin \frac{n\pi x}{l} e^{-b^2 c^2 t}$$

Most general soln is

$$u(x, t) = \sum_{n=1}^{\infty} b_n \sin \frac{n\pi x}{l} \cdot e^{-b^2 c^2 t} \quad (9)$$

(where $C_2 C_3 = b_n$).

Initial condition is $u(x, 0) = u_0$.

$\Rightarrow$ Put $t=0$ in (9)

$$u_0 = \sum_{n=1}^{\infty} b_n \sin \frac{n\pi x}{l} \rightarrow \text{which}$$

represent half-range sine Fourier series.

$$\text{where } b_n = \frac{2}{l} \int_0^l u_0 \cdot \sin \frac{n\pi x}{l} dx$$

$$= \frac{2u_0}{l} \left[-\cos \frac{n\pi x}{l} \right]_0^l$$

$$= \frac{2u_0}{l} \times \frac{l}{n\pi} \left[-\cos \frac{n\pi x}{l} + 1 \right]$$

$$= \frac{2u_0}{n\pi} \left[-\cos n\pi + 1 \right]$$

$$= \frac{240}{n\pi} [1 - (-1)^n]$$

Putting this value in (9)

$$u(x, t) = \sum_{n=1}^{\infty} \frac{240}{n\pi} [1 - (-1)^n] \cdot \sin \frac{n\pi x}{l} e^{-c^2 b^2 t}$$

Ans.

2) Solve the equation $\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}$

with boundary conditions $u(x, 0) = 3 \sin n\pi x$, $u(0, t) = 0$, $u(l, t) = 0$ where $0 < x < l$.

Sol Given $\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}$ (here $c^2 = 1$) — (1)

Boundary conditions $\rightarrow$

$$u(0, t) = 0, \quad u(l, t) = 0. \quad \text{--- (2)}$$

initial condition

$$u(x, 0) = 3 \sin n\pi x. \quad \text{--- (3)}$$

The most general soln is

$$u(x, t) = (C_1 \cos bx + C_2 \sin bx) e^{-b^2 t} \quad \text{--- (4)}$$

when $x = 0$, $u = 0$

$$0 = C_1 \cdot e^{-b^2 t}$$

$\Rightarrow$ i.e. $u = 0$.

$$u(x, t) = C_2 C_3 \sin bx \cdot e^{-b^2 t} \quad \text{--- (5)}$$

when $u(1, t) = 0$ i.e. $x = 1, u = 0$

$$\Rightarrow 0 = C_2 \sin b e^{-b^2 t}$$

Now $C_2 \neq 0$ so $\sin b = 0$

$$\sin b = \sin n\pi$$

$$\Rightarrow b = n\pi$$

$\Rightarrow$ Now eqn (5) becomes

$$u(x, t) = C_2 C_3 \sin n\pi x \cdot e^{-(n\pi)^2 t}$$

$\therefore C_2 C_3 = b_n$ so the most general form is

$$u(x, t) = \sum_{n=1}^{\infty} b_n \cdot e^{-n^2 \pi^2 t} \sin n\pi x \quad \text{--- (6)}$$

when $t = 0, u = 3 \sin n\pi x$

$$\Rightarrow 3 \sin n\pi x = \sum_{n=1}^{\infty} b_n \cdot \sin n\pi x$$

on comparing $b_n = 3$

using in (6)

$$u(x, t) = \sum_{n=1}^{\infty} 3 \sin n\pi x e^{-n^2 \pi^2 t}$$

Ans.